

Upscaling Benthic Habitat Classification from Drone to Satellite Imagery Using Multi-Temporal Machine Learning

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TAKE-HOME MESSAGE

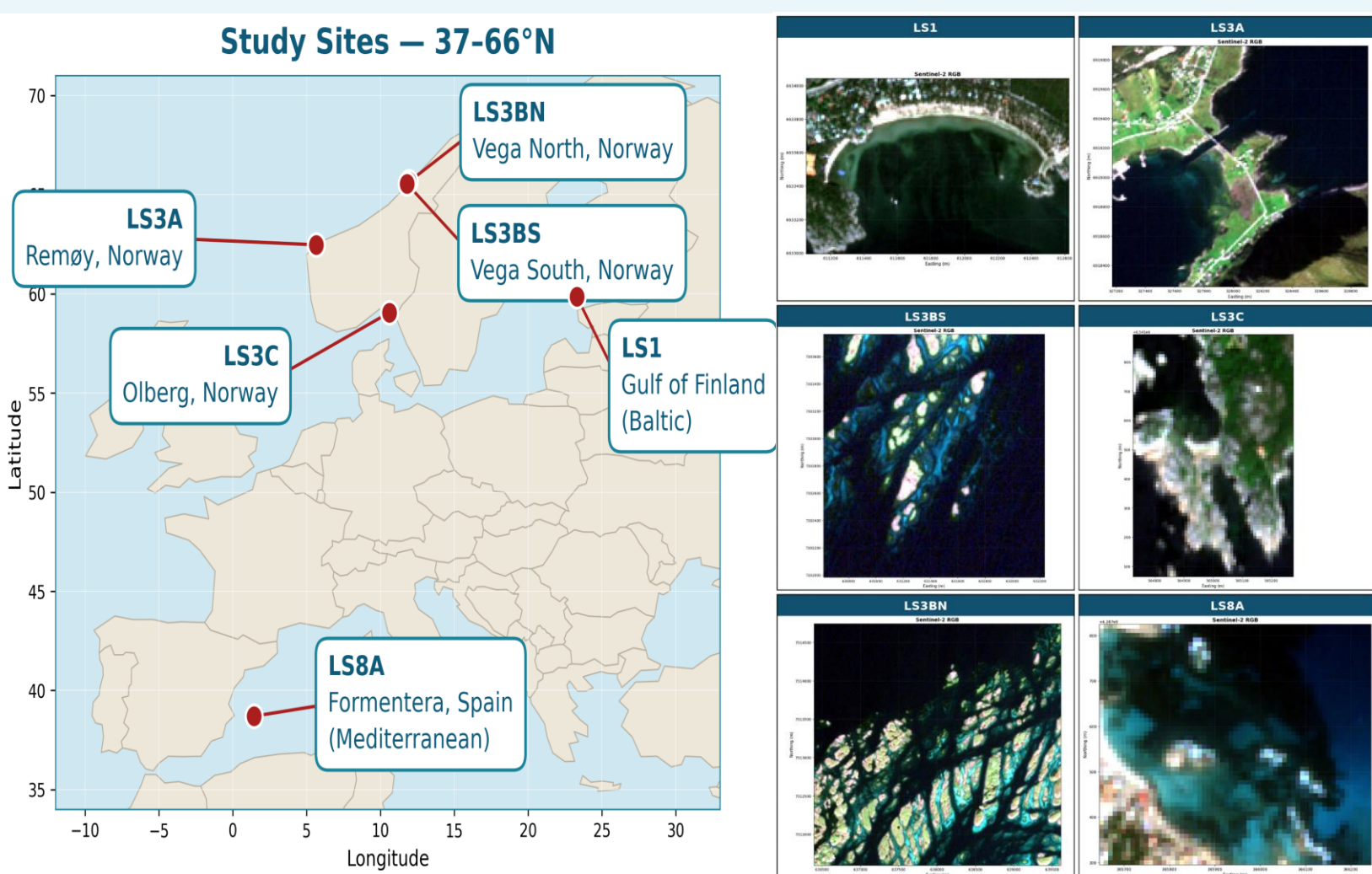
Drone-derived training labels match field-annotation performance for Sentinel-2 benthic habitat mapping and scale reliably within their trained biogeographic range — but extrapolating to entirely new regions remains limited for spectrally distinct classes like seagrass.

1 The Challenge

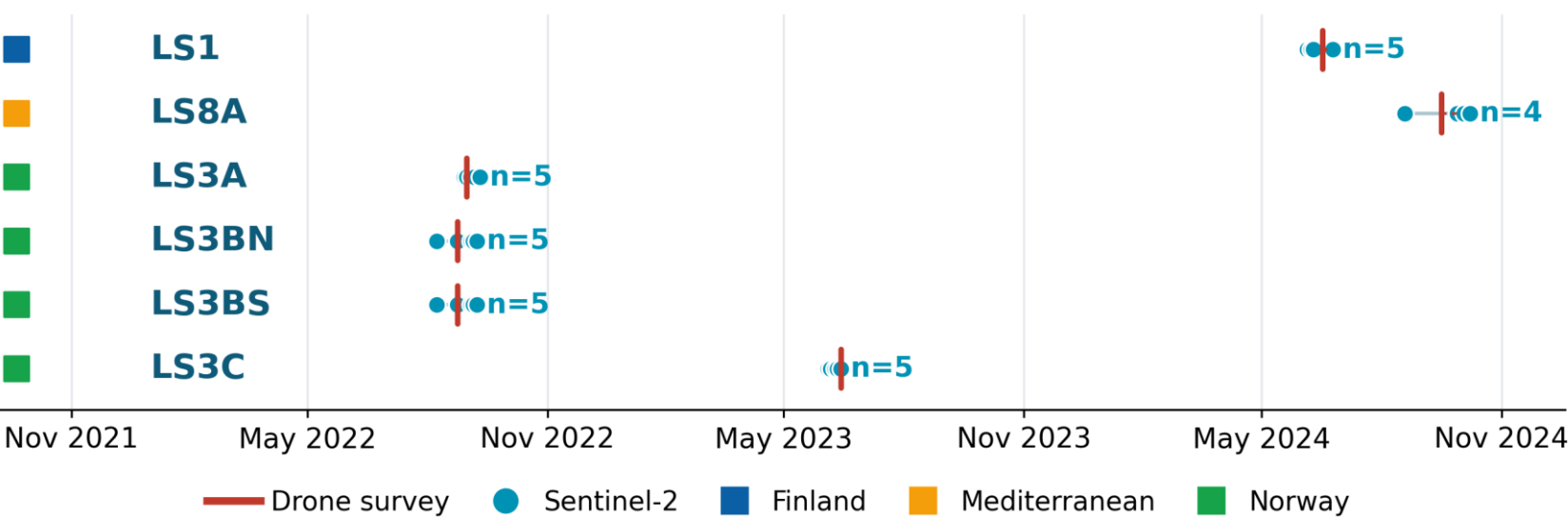
- Shallow coastal habitats — seagrass, macroalgae — are declining fast yet remain poorly monitored at regional scales.
- Satellite mapping scales, but 10 m operational classification needs training strategies that bridge fine-scale field ecology and coarser imagery.
- First systematic, same-footprint comparison of field-annotation vs. drone-derived training labels for Sentinel-2 mapping, across Europe.
- Also demonstrates operational upscaling to a basin-scale area with no local ground truth.

2 Study Sites and data

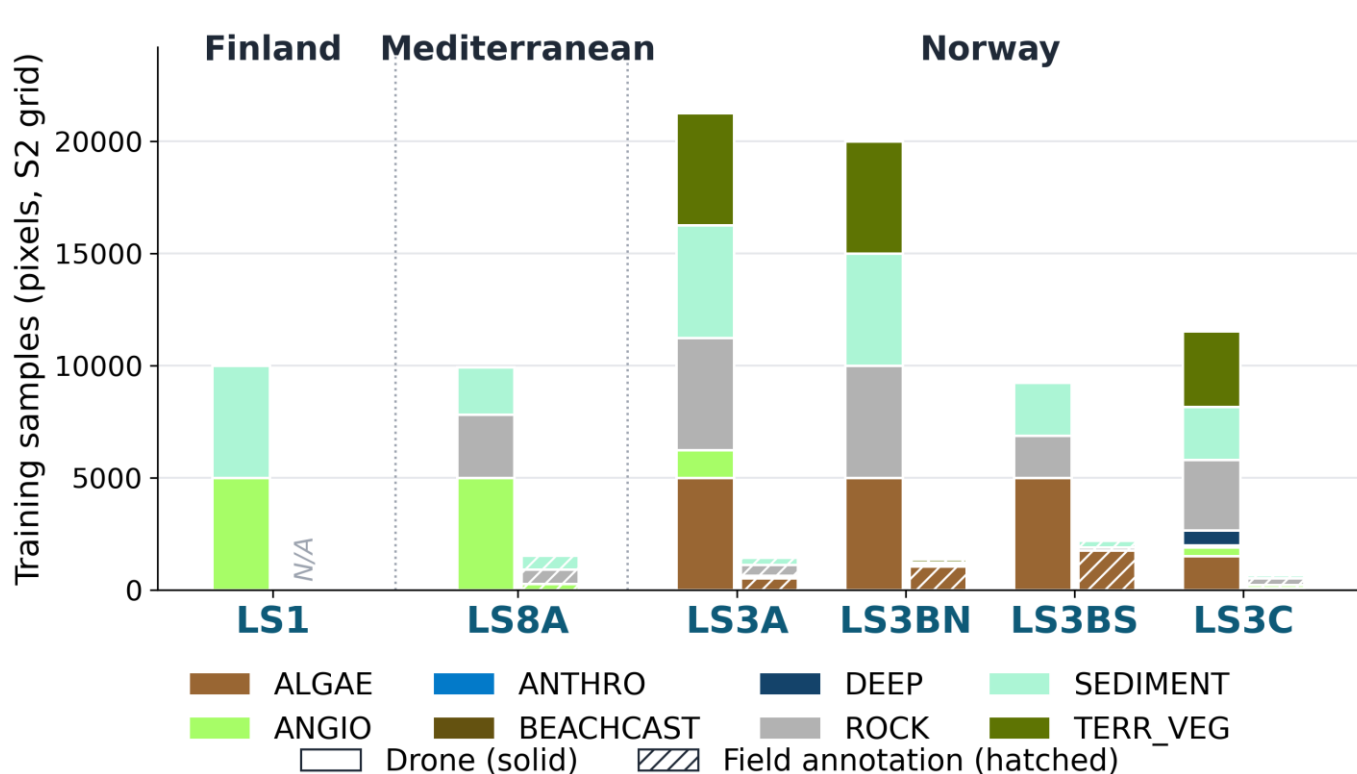
Six biogeographically diverse Learning Sites (37–66°N), each matched to multiple Sentinel-2 acquisitions (29 images, 2022–2024) and resampled into per-site training sets of up to 21,245 pixels.



Temporal alignment: drone surveys and Sentinel-2 acquisitions
29 Sentinel-2 images, 2022–2024



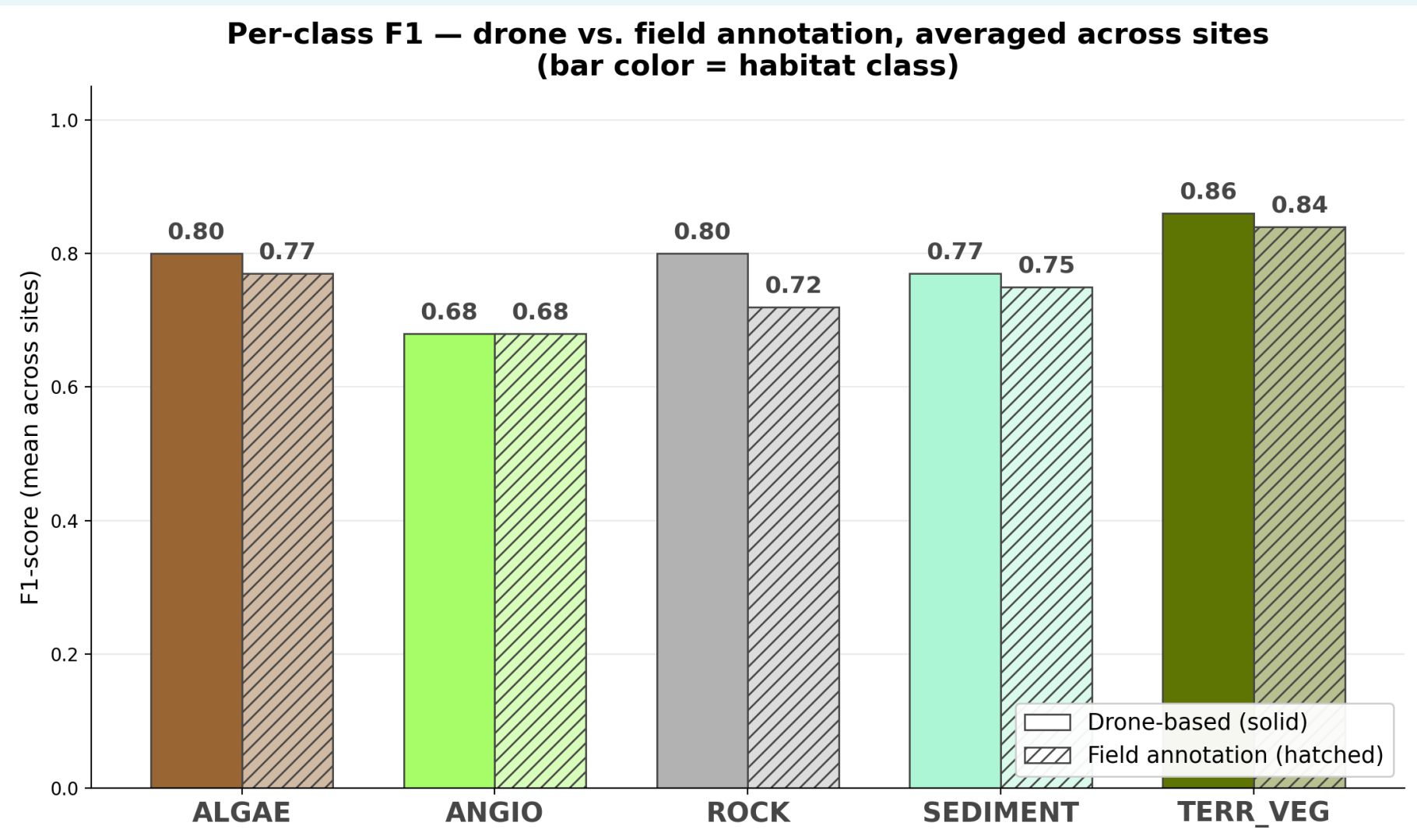
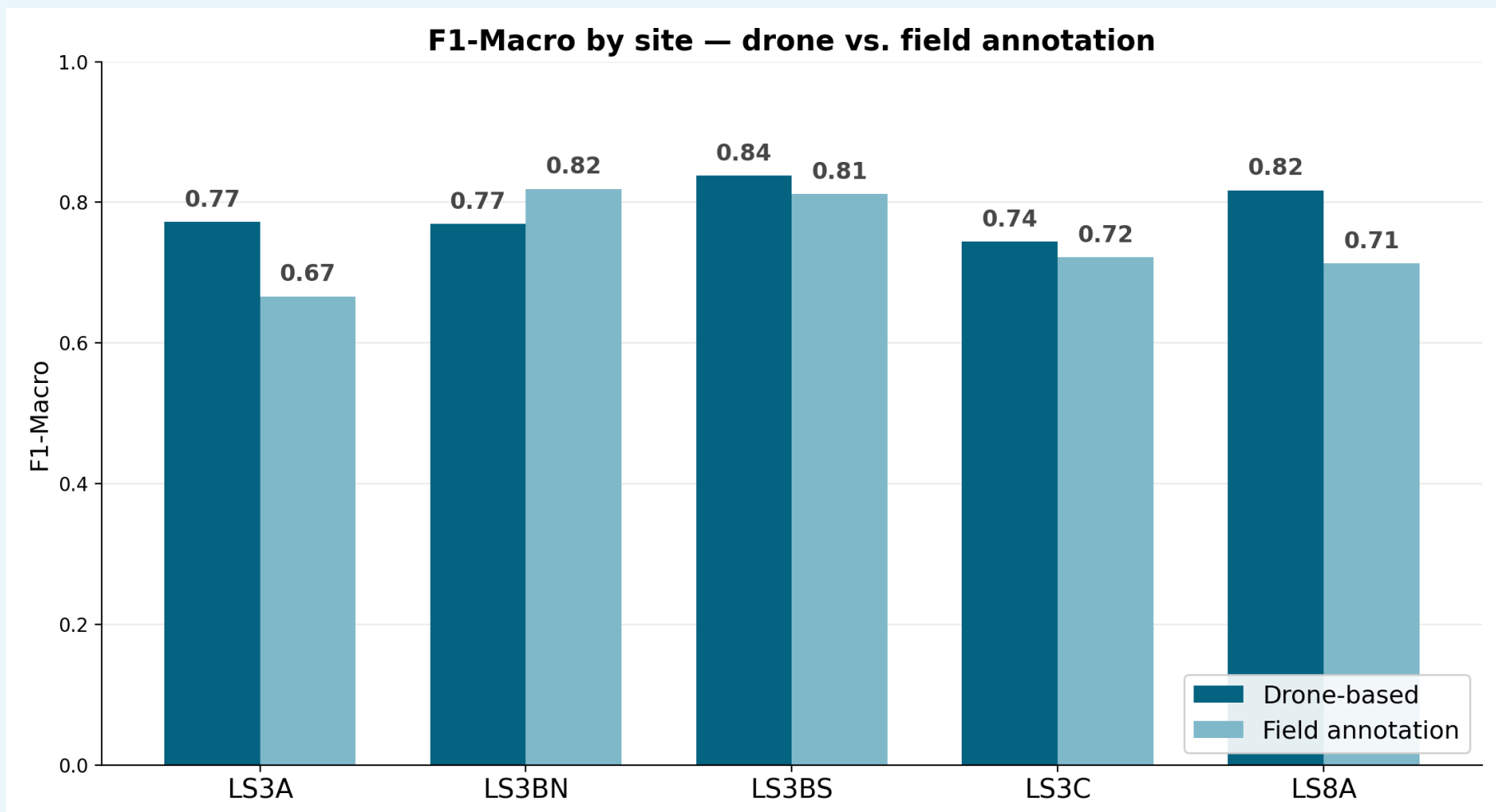
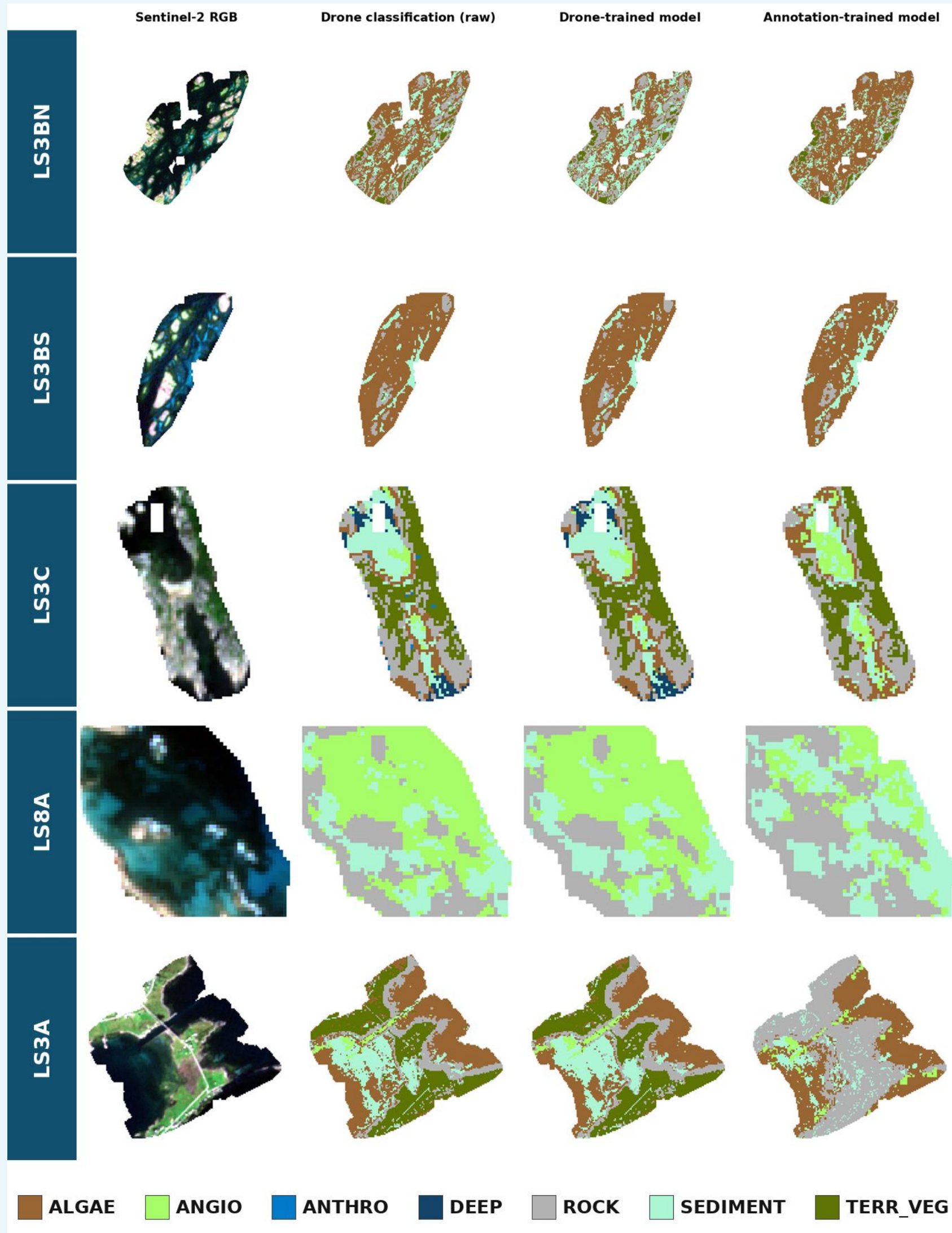
Training samples by site — drone vs. field annotation



4 Drone-Based vs. Field-Annotation Labels

Same footprint, site by site

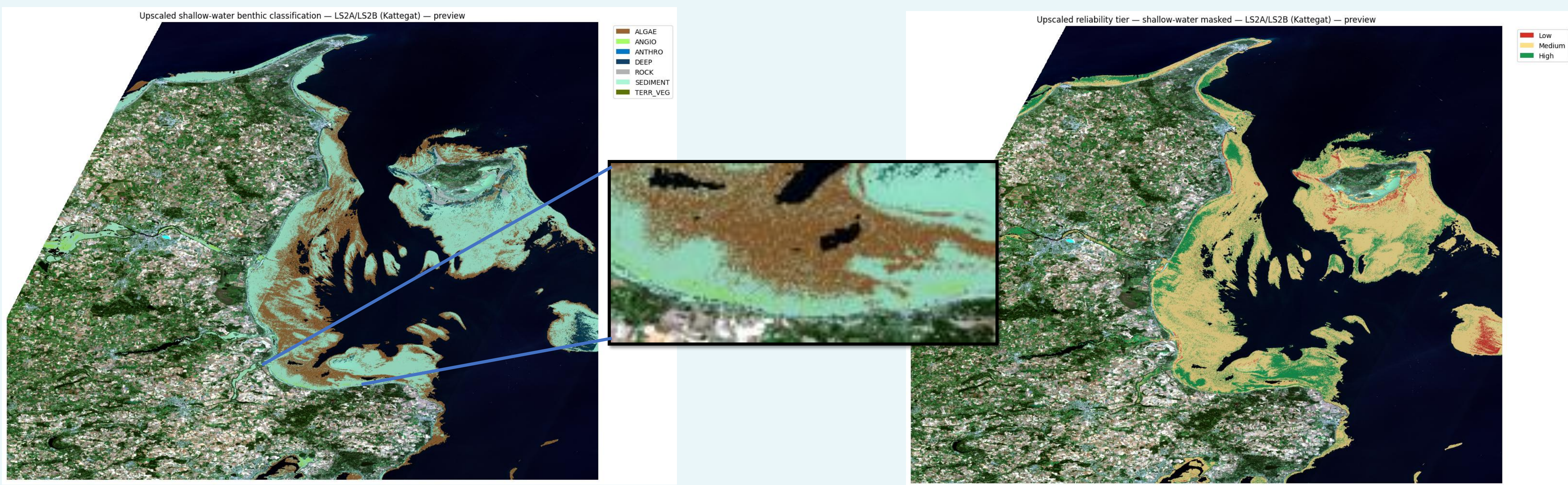
For each of the 5 sites with field annotations, the annotation- and drone-trained models are evaluated on the identical drone-footprint extent, so panels are pixel-for-pixel comparable. The raw drone classification (2nd column) is ground truth, not a model prediction.



5 Operational Application

Basin-scale upscaling to the Kattegat Strait

- Pooled model (all 6 sites) applied to the Kattegat Strait — 382M-pixel Sentinel-2 scene, no local drone ground truth.
- Reliability tiers = pixel-level confidence from prediction entropy — a proxy for trust, not accuracy (no local data to validate against).



3 Method

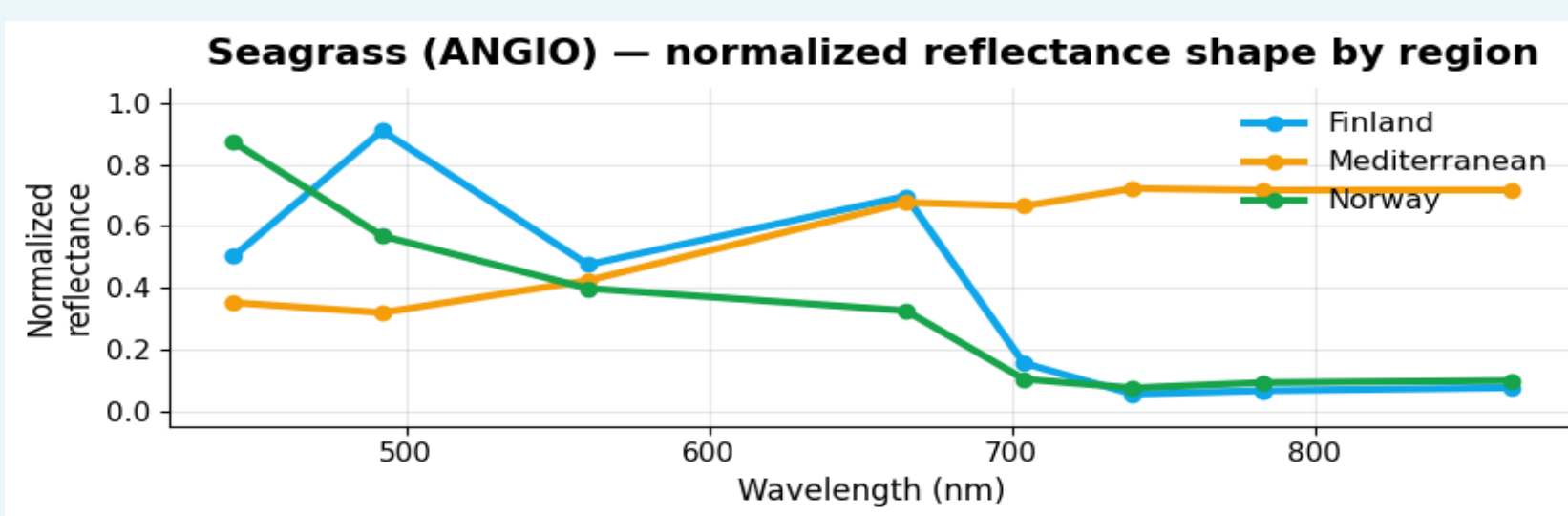
Comparing label sources, then scaling the drone-based model.



6 Generalization Limits

Leave-one-region-out (LORO) stress test

- Each macro-region (Finland / Mediterranean / Norway) was held out entirely from training and tested as if it had no local data.
- Cohen's Kappa drops from 0.62 (pooled model, all 6 regions seen in training) to ≈ 0 when a whole region is unseen — near-random.
- Root cause: seagrass (ANGIO) has a genuinely different spectral signature between regions (right) — not a pipeline artifact. Substrate classes (rock, sediment) stay consistent.



7 Future Work

- Expand training coverage per class across regions and test region-invariant spectral features to close this generalization gap.

REFERENCES

Kuhwald, K., Schneider von Deimling, J., Schubert, P. & Oppelt, N. (2022). How can Sentinel-2 contribute to seagrass mapping in shallow, turbid Baltic Sea waters? Remote Sensing in Ecology and Conservation, 8, 328–346.

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